Lab 16: Population & Systems Ecology (Module 15)

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you should be able to:

1. Interpret **exponential vs logistic** growth from verbal and tabular data, including **where per-capita growth is highest** along an S-curve.
2. Classify **density-dependent** vs **density-independent** regulation and connect each to **management levers** (what you can and cannot move).
3. Assign **trophic roles** and trace **energy** and **nutrients** through an ecosystem, including decomposers.
4. Apply the **~10% energy transfer** rule across multiple steps and explain **food-chain length** and **human food choices** in those terms.

Overview

You are the **acting manager** of the fictional **Green Mesa National Wildlife Refuge**. Elk were reintroduced in 2010; your field team sends **population notes** and **energy budgets** (invented but numerically consistent). You will **forecast** growth, **diagnose** regulation, compare **trophic "pyramids,"** and **audit** energy through a food web—work a reserve team might use to set harvest, fire, and habitat priorities.

Materials Needed

- Pencil or pen
- This worksheet
- Calculator (optional)

Part 1: Elk Come Home — From Boom to Ceiling

Elk counts (made-up):

Year	Adult elk	Notes
2012	40	Few predators; abundant browse
2014	90	Still rising fast
2016	155	Growth slowing
2018	198	Near stable
2020	205	Fluctuates ±5

Mean numerical increase per interval (adults/year): 2012→2014: +25/yr; 2014→2016: +32.5/yr; 2016→2018: +21.5/yr; 2018→2020: +3.5/yr (use if helpful).

1. Shape of growth: Which curve best fits 2012–2018—**exponential (J)** or **logistic (S)**? **Circle one** and cite **two** observations from the table (or the Δ line) that support your choice.

2. Where growth is "fastest" in an S-curve: On a logistic curve, **per-capita** growth rate is typically **highest** when the population is **well below** *K* and **lower** as *N* approaches *K*. Does the 2012–2018 pattern match that story—**yes / no**, and **one** sentence of support using years or numbers.

3. Carrying capacity: From the table, estimate **K** (a range is fine, e.g., "about ____ elk"). What **resource** is most likely limiting elk if the story continued? Name **one** management action that could **raise** *K* for elk without changing the climate (short phrase).

- Estimated *K*:
- Limiting resource (hypothesis):
- One way to **increase** *K* (habitat or food side):

4. Prediction: If managers **remove half the elk** in 2021 and browse recovers, what do you predict about **per-capita growth rate** the next few years—**increase / decrease / stay same**? Tie your answer to N relative to K in **one** sentence.

Part 2: Regulators, Management, and the Trophic Ladder

Three events; same big ideas: who limits the herd and where energy goes.

Event A: A **parasitic fly** spreads; infection probability rises as elk crowd into the same valleys.
Event B: A **wildfire** burns 40% of the refuge regardless of elk density.
Event C (new plant invader): A non-native shrub **outcompetes** bunchgrass; total **forage** for elk drops until crews remove the shrub, even at low elk numbers.

5. Classify each as **density-dependent**, **density-independent**, or **mainly a resource/ food limit** (choose the **best** label; some cases blur—defend in one short phrase if you need to).

Event	Type (best label)	One reason
Event A		
Event B		
Event C		

6. Management puzzle: For **which** event is **herd size** a useful dial for reducing harm (alongside other work)? For **which** is the main tool **not** "shoot fewer or more elk" but **fire / vegetation / invasives**?

- Herd size matters most for:
- Mostly habitat/weather/invasion tools:

7. Why does a given producer energy budget generally support less predator biomass than herbivore biomass? Explain energy transfers in **2–3 sentences**. Can energy alone predict exact animal counts without knowing body sizes and energy needs?

Bridge: **Trophic levels** connect **who limits whom** (Part 2) to **how much energy** reaches the top of the food web (Part 4).

Part 3: Web Assembly — Who Eats Whom?

Refuge food web (simplified):

- **Sunlight** → bunchgrass & shrubs (**producers**)
- **Grasshoppers** eat plants
- **Ground-feeding birds** eat grasshoppers
- **Hawks** eat birds
- **Coyotes** eat elk calves (sometimes) and scavenge carrion
- **Fungi & bacteria** decompose dead material

8. Build one linear chain with **four trophic levels** (producer → … → top carnivore). Use arrows **from food to eater** (e.g. grass → grasshopper → ground-feeding bird → hawk).

9. Label roles for the chain you wrote (producer, primary consumer, secondary, tertiary). If **coyotes** are in a **different** chain than your hawk chain, name **coyote's** most likely **trophic** role here (**secondary** or **tertiary** consumer vs scavenger) in one line.

10. Pyramids (concept check): A **pyramid of numbers** can look "wrong" (many small herbivores on one big tree) while a **pyramid of energy transfer per unit area per unit time** is **upright** for the same system. In **2–3** sentences, explain **why** energy pyramids are the usual standard for "how much life can a habitat support at each level."

11. Nutrients vs energy: In **2–3** sentences, explain why **decomposers** matter to **producers** even though energy does **not** cycle from dead matter back to sunlight.

Part 4: Energy Audit — The ~10% Rule

Use ~10% transfer between trophic levels (textbook-style rounding). At each step, assume one-tenth of the energy stored as new biomass at one level becomes new biomass at the next level. This is a simplifying model; real efficiencies vary. Show your calculation at each step.

Given: Producers store **1,000,000 kJ** as new plant biomass in one season. Use this producer-level budget as the starting point. Apply the assumed **10% transfer once per trophic step**, including the producer-to-herbivore step.

12. Step the energy down the chain:

Step	Available energy (kJ)
To primary consumers (herbivores)	
To secondary consumers	
To tertiary consumers	

13. Compare the budgets. Divide the tertiary-consumer budget by the primary-consumer budget. How many kJ of tertiary-consumer biomass does this model yield for every **1 kJ** of primary-consumer biomass? State the ratio and the number of intervening transfers.

14. Sanity check: If humans tried to feed the world only on **apex tuna**, would we need **more** or **less** ocean primary production than if we ate **plants** (or **herbivorous** fish) directly? Use **2–3** sentences and the table logic, not a moral judgment about diet.

15. One-sentence contrast:

- **Energy through ecosystems:**

- **Nutrients through ecosystems:**